

**Project Design Phase-II**  
**Data Flow Diagram & User Stories**

|               |                                                     |
|---------------|-----------------------------------------------------|
| Date          | 29 June 2026                                        |
| Name          | HARRI VENDHAN L                                     |
| Project Name  | <b>Metro Ticket Generating System in ServiceNow</b> |
| Maximum Marks | 4 Marks                                             |

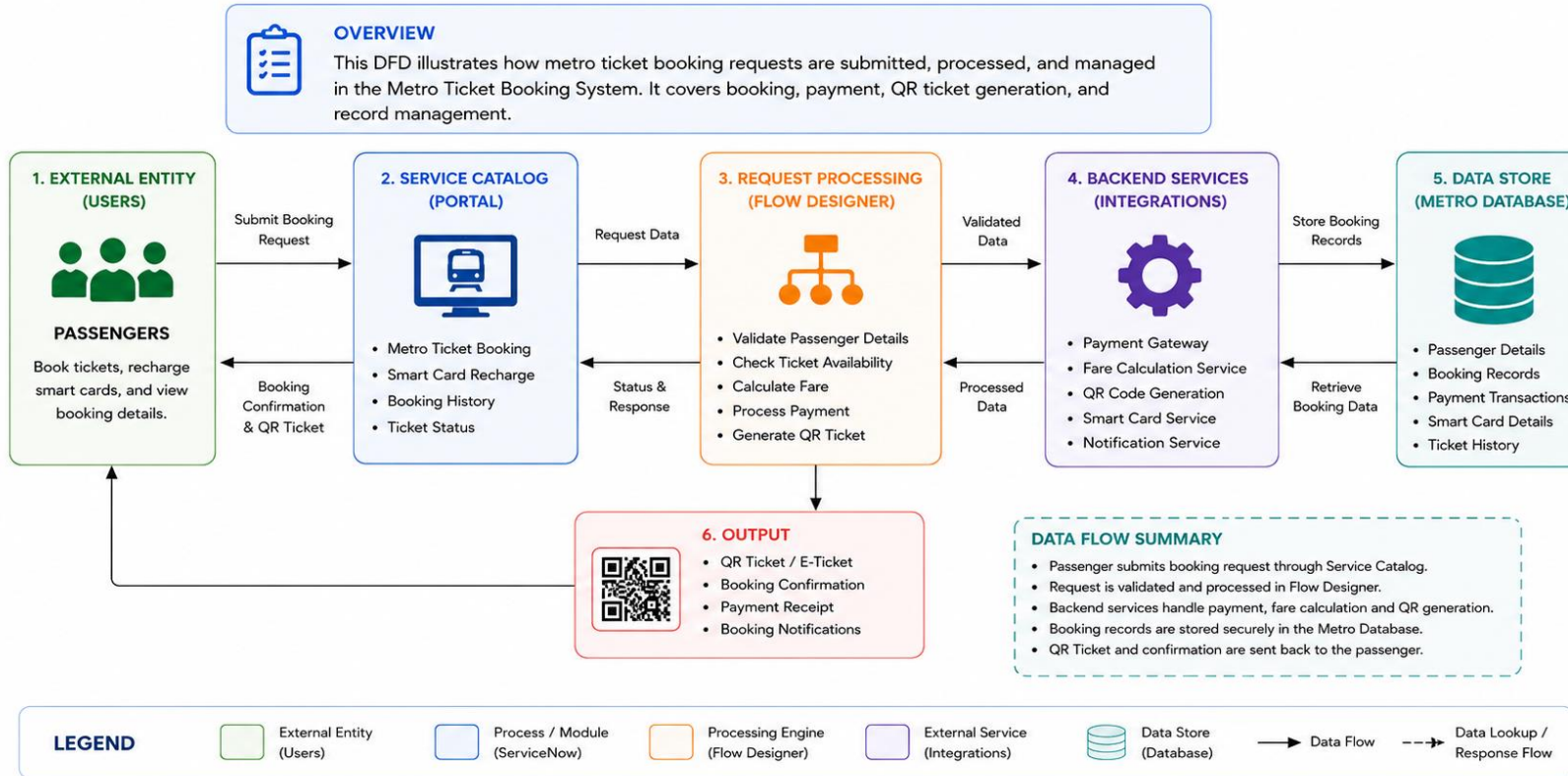
**Data Flow Diagrams:**

A **Data Flow Diagram (DFD)** illustrates the movement of information within the **Metro Ticket Booking System** developed in **ServiceNow**. It shows how passenger details and booking requests are submitted through the Service Catalog, validated by the Flow Designer, processed for fare calculation and QR code generation, and finally stored in the **Metro Database**. The diagram also demonstrates how the generated QR ticket is made available to the passenger for seamless metro travel while ensuring secure and efficient management of booking records.

**Example:** [\(Simplified\)](#)

## DATA FLOW DIAGRAM

### Metro Ticket Booking System (ServiceNow)



## User Stories

Use the below template to list all the user stories for the product.

## User Stories

| User Type                                                                                               | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                                                        | Acceptance Criteria                                                                                                                                                                                                | Priority | Release                     |
|---------------------------------------------------------------------------------------------------------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|
|  Passenger (Customer)  | Metro Ticket Booking          | MTB-01            | As a Passenger, I can search for metro routes, select source and destination stations, choose travel date and time, and book a ticket so that I can travel conveniently. | <ul style="list-style-type: none"> <li>Search routes and view available trains</li> <li>Select date and time</li> <li>Book a ticket successfully</li> <li>QR ticket is generated and confirmed</li> </ul>          | Critical | Phase 1 (Core)              |
|  Passenger (Customer)  | Smart Card Recharge           | MTB-02            | As a Passenger, I can recharge my smart card online so that I can use it for metro travel.                                                                               | <ul style="list-style-type: none"> <li>Enter smart card details</li> <li>View current balance</li> <li>Make payment and recharge</li> <li>Updated balance is reflected instantly</li> </ul>                        | High     | Phase 1 (Core)              |
|  Passenger (Customer)  | View Booking History          | MTB-03            | As a Passenger, I can view my booking history and past transactions so that I can keep track of my travel records.                                                       | <ul style="list-style-type: none"> <li>View all past bookings</li> <li>View smart card recharges</li> <li>View payments with date, time, amount</li> <li>View booking status</li> </ul>                            | Medium   | Phase 2 (Enhanced Features) |
|  Passenger (Customer)  | Cancel / Refund Ticket        | MTB-04            | As a Passenger, I can cancel my ticket and request a refund (if eligible) so that I can get my money back.                                                               | <ul style="list-style-type: none"> <li>Cancel eligible bookings</li> <li>Validate refund rules</li> <li>Process refund</li> <li>Send cancellation confirmation</li> </ul>                                          | Medium   | Phase 2 (Enhanced Features) |
|  Passenger (Customer)  | Receive Notifications         | MTB-05            | As a Passenger, I want to receive notifications for booking confirmation, payment, and travel updates.                                                                   | <ul style="list-style-type: none"> <li>Send email/SMS notifications</li> <li>Booking confirmation</li> <li>Payment status updates</li> <li>Travel alerts and metro updates</li> </ul>                              | High     | Phase 1 (Core)              |
|  Admin (Metro Admin) | Manage Routes & Fares         | MTB-06            | As an Administrator, I can manage metro routes, stations, and fares so that accurate information is available to passengers.                                             | <ul style="list-style-type: none"> <li>Add / Update / Deactivate routes &amp; stations</li> <li>Set and update fare rules</li> <li>Changes reflected in booking system</li> <li>Validate data integrity</li> </ul> | High     | Phase 2 (Admin Features)    |
|  Admin (Metro Admin) | Generate Reports              | MTB-07            | As an Administrator, I can generate reports for bookings, revenue, and smart card usage so that I can analyze performance.                                               | <ul style="list-style-type: none"> <li>Daily booking reports</li> <li>Revenue summary reports</li> <li>Smart card recharge reports</li> <li>Export reports with filters</li> </ul>                                 | Medium   | Phase 3 (Extended Features) |